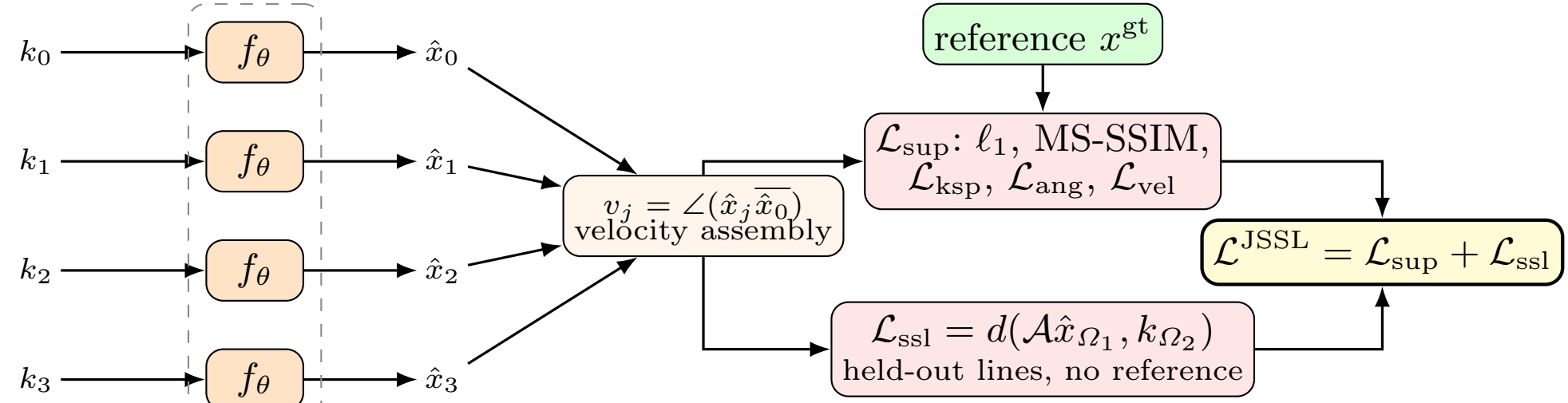

k_0
k_1
k_2
k_3
f_θ
f_θ
f_θ
f_θ
$\hat{x}_0$
$\hat{x}_1$
$\hat{x}_2$
$\hat{x}_3$
one shared θ, four separate passes; encodings never meet here
$v_j = \angle(\hat{x}_j \overline{\hat{x}_0})$ velocity assembly
reference x^{gt}
$\mathcal{L}_{\mathrm{sup}}$: ℓ_1, MS-SSIM, $\mathcal{L}_{\mathrm{ksp}}$, $\mathcal{L}_{\mathrm{ang}}$, $\mathcal{L}_{\mathrm{vel}}$
$\mathcal{L}_{\mathrm{ssl}} = d(\mathcal{A}\hat{x}_{\Omega_1}, k_{\Omega_2})$ held-out lines, no reference
$\mathcal{L}^{\mathrm{JSSL}} = \mathcal{L}_{\mathrm{sup}} + \mathcal{L}_{\mathrm{ssl}}$